

# ● Computer Organization | RISC and CISC

A microprocessor is a processing unit on a single chip. It is an integrated circuit that performs the core functions of a computer CPU. It is a multipurpose programmable silicon chip constructed using Metal Oxide Semiconductor (MOS) technology which is clock-driven and register-based. It accepts binary data as input and provides output after processing it as per the specification of instructions stored in the memory. These microprocessors are capable of processing 128 bits at a time at the speed of one billion instructions per second.

## **Characteristics of a microprocessor:**

### **Instruction Set –**

The set of complete instructions that the microprocessor executes is termed the instruction set.

### **Word Length –**

The number of bits processed in a single instruction is called word length or word size. Greater the word size, the larger the processing power of the CPU.

### **System Clock Speed –**

Clock speed determines how fast a single instruction can be executed in a processor. The microprocessor's pace is controlled by the System Clock. Clock speeds are generally measured in millions of cycles per second (MHz) and thousand million cycles per second (GHz). Clock speed is considered to be a very important aspect of predicting the performance of a processor.

## **Classification of Microprocessors:**

Besides the classification based on the word length, the classification is also based on the architecture i.e. Instruction Set of the microprocessor. These are categorized into RISC and CISC.

### **Reduced Instruction Set Architecture (RISC) –**

The main idea behind this is to make hardware simpler by using an instruction set composed of a few basic steps for loading, evaluating, and storing operations just like a load command will load data, a store command will store the data.

### **Complex Instruction Set Architecture (CISC) –**

The main idea is that a single instruction will do all loading, evaluating, and storing operations just like a multiplication command will do stuff like loading data, evaluating, and storing it, hence it's complex.

Both approaches try to increase the CPU performance

- **RISC:** Reduce the cycles per instruction at the cost of the number of instructions per program.
- **CISC:** The CISC approach attempts to minimize the number of instructions per program but at the cost of an increase in the number of cycles per instruction.

$$CPU\ Time = \frac{Seconds}{Program} = \frac{Instructions}{Program} \times \frac{Cycles}{Instructions} \times \frac{Seconds}{Cycle}$$

Earlier when programming was done using assembly language, a need was felt to make instruction do more tasks because programming in assembly was tedious and error-prone due to which CISC architecture evolved but with the up rise of high-level language dependency on assembly reduced RISC architecture prevailed.

### Characteristic of RISC –

1. Simpler instruction, hence simple instruction decoding.
2. Instruction comes undersize of one word.
3. Instruction takes a single clock cycle to get executed.
4. More general-purpose registers.
5. Simple Addressing Modes.
6. Fewer Data types.
7. A pipeline can be achieved.

### Characteristic of CISC –

1. Complex instruction, hence complex instruction decoding.
2. Instructions are larger than one-word size.
3. Instruction may take more than a single clock cycle to get executed.
4. Less number of general-purpose registers as operations get performed in memory itself.
5. Complex Addressing Modes.
6. More Data types.

**Example –** Suppose we have to add two 8-bit numbers:

- **CISC approach:** There will be a single command or instruction for this like ADD which will perform the task.
- **RISC approach:** Here programmer will write the first load command to load data in registers then it will use a suitable operator and then it will store the result in the desired location.

So, add operation is divided into parts i.e. load, operate, store due to which RISC programs are longer and require more memory to get stored but require fewer transistors due to less complex command.

## Difference –

| <b>RISC</b>                                                 | <b>CISC</b>                                           |
|-------------------------------------------------------------|-------------------------------------------------------|
| Focus on software                                           | Focus on hardware                                     |
| Uses only Hardwired control unit                            | Uses both hardwired and microprogrammed control unit  |
| Transistors are used for more registers                     | Transistors are used for storing complex Instructions |
| Fixed sized instructions                                    | Variable sized instructions                           |
| Can perform only Register to Register Arithmetic operations | Can perform REG to REG or REG to MEM or MEM to MEM    |
| Requires more number of registers                           | Requires less number of registers                     |
| Code size is large                                          | Code size is small                                    |
| An instruction executed in a single clock cycle             | Instruction takes more than one clock cycle           |
| An instruction fit in one word                              | Instructions are larger than the size of one word     |